MILITARY SPECIFICATION

MOTOR-GENERATORS, DC TO AC, SHIPBOARD SERVICE

This amendment forms a part of Military Specification MIL-M-19097B(SHIPS), dated 22 May 1961, and is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

PAGES 1 and 2

* 2.1, under "SPECIFICATIONS, MILITARY": Delete reference to "MIL-D-963", "MIL-C-2174", "MIL-G-3545", "MIL-L-15719", "MIL-P-16018", "MIL-I-17166", "MIL-R-18281", and "MIL-E-22843", and add the following:

"MIL-C-2212 - Controllers, Electric Motor, AC or DC, and Associated Switching Devices, Naval Shipboard.
DOD-D-1000 - Drawings, Engineering and Associated Lists.
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts, Packaging of.
MIL-G-18709 - Grease, Ball and Roller Bearing.
MIL-I-24137 - Iron Castings; Nodular Graphitic (Ductile Iron) and Nodular Graphitic (Corrosion Resisting, Austenitic, Low Magnetic Permeability) (For Submarine Application)."

PAGE 2

* 2.1, under "STANDARDS, MILITARY": Add:

"MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria."

* 2.2: Delete reference to "OFFICIAL CLASSIFICATION COMMITTEE" and substitute the following:

"UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)"

FSC 6125

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* 2.2: Add:

"AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A 47 - Malleable Iron Castings. (DOD Adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)"

PAGE 3

3.2.8: Delete and substitute:

"3.2.8 Ambient temperature. The motor generators and all associated control equipment shall be designed for operation in a maximum ambient temperature of 50 degrees Celsius (^{o}C) and shall also meet the performance requirements specified herein down to 0°C."

PAGE 5

3.2.27: Delete and substitute:

"3.2.27 System short circuit performance. The motor generator, under automatic control, shall produce short circuit current to satisfy the following requirements:

(a) $I^2t = 1100$

Where:
I = Sustained steady state generator output current (per unit valve).
t = Time in seconds.

(b) The minimum value of generator short circuit current shall be three per unit.

(c) The peak value of short circuit current shall not exceed 11 per unit. The requirements of (a) and (b) herein shall be met under the following three phase short circuit condition for three-phase generators or one-phase short circuit condition for one-phase generators:

Initial load	Control	Generator field temperature
No load	Automatic	Hot

Hot field temperature shall be equivalent to normal full load rise obtained from the heat run plus the specified maximum ambient temperature (50°C). The design of the motor generators and control equipment shall be such that it can withstand a short circuit current of the magnitude required without damaging the equipment. Upon removal of the short circuit, rated voltage shall be automatically restored."

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3.2.28, item (a), line 2: Delete "number of" and substitute "three", and add the following as item (b):

"(b) The generators shall be paralleled 180 degrees out of phase without damage to the equipment."

PAGES 6 and 7

3.3.12, lines 4 and 5: Delete "Specification MIL-E-22843" and substitute "MIL-STD-740"; item (a), line 3: Delete "Grade C" and substitute "Grade D".

PAGE 7

* Table III, under "Name of part" column: Delete "Coils" and substitute "Coils and coil connectors".

3.3.13, lines 4 and 5: Delete "Specification MIL-E-22843" and substitute "type 3 of MIL-STD-740"; lines 6 and 7, delete "Grade - Grade B or C equipment as specified (see 6.1)".

* Add as new paragraph 3.3.13.1:

"3.3.13.1 In designing for low structureborne noise, factors to be considered shall include, but not be limited to the following:

(a) Precision machining of rolling, rubbing and fitted parts.
(b) Construction and fabrication such that minimum amount of correction is necessary for balancing.
(c) Avoiding sharp cutoffs and turbulence in air cooling systems.
(d) Number of slots per pole pitch.
(e) Number of slots magnetically under one pole.
(f) Slot frequency as a function of the natural frequency of the magnetic frame.
(g) Skewing of stator slots.
(h) Skewing of rotor slots."

PAGE 8

* Table IV: Delete and substitute:

TABLE IV. Minimum material requirements.

Item	Material	Specification	Limitations
Brush Caps and cartridge Ball bearing end brackets, terminal boxes, covers, frames, flanges, vee rings, shells, drip covers, baffles, hand hole as access covers	Various Steel Malleable iron Nodular graphitic iron	MIL-B-3743 See Note 1 ASTM A 47 MIL-I-24137	See MIL-B-3743

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TABLE IV. Minimum material requirements. - Continued

Item	Material	Specification	Limitations
Eye bolts	Steel		
Fans	Steel Aluminum	1/	
Grease	Petroleum	MIL-G-18709	Bearings operating at 110°C and below
	Silicone	MIL-G-18709	Bearings operating at total temperature above 110°C
Holders, brush bodies, frames, fingers	Brass Bronze Steel		Corrosion resistant or treated for corrosion resistance
Springs	Steel, carbon		Corrosion resistant or treated for corrosion resistance
	Steel corrosion resistant Copper-beryllium		
Shafts	Steel		Tensile strength not less than 75,000 lb/in^2 and elongation not less than 20 percent in 2 inches
Spider and quills	Steel	1/	
Studs, brush holder	Steel	1/	Treated for corrosion resistance
	Brass	Commercial	

1/ Unless otherwise indicated in table IV, steel parts may be cast, fabricated, wrought or forged. Cast steel shall be in accordance with class B of MIL-S-15083, or equivalent, except that radiographic and magnetic particle inspection will not be required."

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3.4.5, line 6: Delete "400" and substitute "60".

PAGE 10

3.4.7.4: Delete and substitute:

"3.4.7.4 Mounting. The bearings shall be mounted as follows:

(a) Fixed free method. When the bearings are secured to the shaft by means of locknuts and lockwashers, the following requirements shall apply:
(1) One end bearing shall be fixed axially in the housing by suitable housing and end cap shoulders. The axial movements of the shaft shall be not more than 0.010 inch for size 310 and smaller bearings and 0.015 inch for bearings larger than size 310.

(b) Opposed shoulder method. When the opposed shoulder method is used, the following information shall be submitted to the contracting activity for approval:
(1) The initial end play required (to compensate for thermal expansion differential between shaft and housing).
(2) The method used to measure end play."

PAGES 10 and 11

3.4.8, last line: Delete "and" and substitute "end".

PAGE 12

3.4.13.4: Add: "Where constant pressure springs are used they shall be such that the brush pressure parallel to the length of the brush will be automatically maintained within 95 to 105 percent of average value throughout the useful life of the brush."

3.4.14, line 11: Delete "translucent" and substitute "transparent".

PAGE 13

3.4.18, third and fourth sentences: Delete.

3.4.19, last sentence: Delete and substitute: "All motor field coil connections shall be bolted, brazed or soldered connections."

3.4.22, item (b): Delete.

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3.5.2.1: Add new item (q):

"(q) Wattmeter for generator output."

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* 3.5.4.7, line 4: Delete "MIL-C-2174" and substitute "MIL-C-2212".

3.5.4.7: Add new item (j):

"(j) The number of resistance steps in the motor controller shall be sufficient to limit the inrush currents to 150 percent rated upon starting."

3.5.4.8: Add new items (f) and (g):

"(f) Overvoltage limitation shall be provided as a part of the voltage regulator to protect the equipment against overvoltage due to loss of sensing or failure of components in the voltage regulator. The overvoltage limiting circuit shall sense the generator terminal voltage and limit it by controlling the last stage of the exciter, should an overvoltage occur. The circuit shall limit the overvoltages between 110 and 130 percent rated. The motor generator shall meet all the requirements of this specification with the overvoltage limit set at any point within its range. The equipment shall not be damaged by overvoltage of 130 percent rated.

(g) The voltage regulator shall be switched from `automatic' to `manual' or from `manual' to `automatic' while the motor generator is in operation without damage to any components. The motor generator shall be started in manual or in automatic."

PAGE 18

3.5.4.9: Add new item (f):

"(f) The frequency regulator shall be switched from `automatic' to `manual' or from `manual' to `automatic' while the motor generator is in operation without damage to any components. The motor generator shall be started in manual or in automatic."

3.5.4.12: Add: "Voltage and frequency adjusting rheostats shall be connected so that turning clockwise increases voltage or frequency and turning counterclockwise decreases voltage or frequency."

* 3.5.4.14.2: Delete.

PAGE 19

3.5.4.20, line 6: Delete "motor" and substitute "meter" and line 9, delete "115" and substitute "120".

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PAGE 20

3.5.4.22: Add: "The meter full scale range shall be selected such that rated amperes fall within approximately 60 to 80 percent of the full scale range."

PAGE 21

3.7.1, line 2: Delete "MIL-D-963" and substitute "DOD-D-1000".

3.7.2, line 2: Delete "MIL-D-963" and substitute "DOD-D-1000".

3.7.2.2, line 4: Delete "MIL-D-963" and substitute "DOD-D-1000".

PAGE 24

* 3.7.3.2, line 6: Delete "MIL-C-2174" and substitute "MIL-C-2212".

PAGE 28

3.7.4, second sentence: Delete and substitute: "The dimensions of the technical manual shall be 11-5/8 inches high and 8-3/4 inches wide. Technical manuals shall conform to type I of MIL-M-15071 and as follows:".

* 4.1: Delete and substitute:

"4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements."

PAGE 29

4.2.1.1, line 13: Delete "Specification MIL-E-22843" and substitute "MIL-STD-740".

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* Table X: Delete and substitute:

"TABLE X. Routine and periodic tests.

Description of test	Applicable test paragraph		Test to be conducted while motor-generators and control equipment are operating together as a complete system	Test which may be conducted on motor-generator alone	Test which may be conducted on control equipment alone
	Routine tests	Periodic tests			
Creepage and clearance	4.6.1	4.6.1		X	X
Enclosure	4.6.2	4.6.2		X	X
Weight		4.6.3		X	X
Air gap	4.6.4	4.6.4		X	
Resistance at 25°C	4.6.5	4.6.5		X	
Insulation resistance (cold)	4.6.6	4.6.6		X	X
Dielectric	4.6.7	4.6.7		X	X
Airborne noise:					
Type I motor generator	4.6.8.1.2	4.6.8.1.1		X	
Type II motor generator	4.6.8.2.2	4.6.8.2.1	X		
Commutation	4.6.9	4.6.9	X		
End play	4.6.10	4.6.10		X	
Inclined operation		4.6.11		X	
Overspeed	4.6.12	4.6.12		X	
Structureborne noise, internally excited:					
Type I motor generator	4.6.13.1.1	4.6.13.1.1		X	
Type II motor generator	4.6.13.2.2	4.6.13.2.1	X		
Externally excited vibration		4.6.14	X		
Inherent speed (frequency) regulation		4.6.15	X		
No load saturation curve for generator		4.6.16		X	
Synchronous impedance curve		4.6.17		X	
Shock		4.6.18	X		
Regulated frequency adjustment range and steps		4.6.19	X		
Regulated voltage adjustment range and steps		4.6.20	X		
Manual frequency adjustment range and steps		4.6.21	X		

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"TABLE X. Routine and periodic tests. - Continued

Description of test	Applicable test paragraph		Test to be conducted while motor-generators and control equipment are operating together as a complete system	Test which may be conducted on motor-generator alone	Test which may be conducted on control equipment alone
	Routine tests	Periodic tests			
Manual voltage adjustment range and steps		4.6.22	X		
Automatic frequency and voltage regulation	4.6.23.2	4.6.23.1	X		
Transient response of frequency regulator (deviation and recovery time)	4.6.24	4.6.24	X		
Transient response of voltage regulator (deviation and recovery time)	4.6.25	4.6.25	X		
Parallel operation (kw and kvar sharing)		4.6.26	X		
Short circuit characteristics		4.6.27	X		
Overspeed protection: Type II motor generator	4.6.28	4.6.28	X		
Rated load heating		4.6.29.2	X		
Special heating tests: Type II motor generator (only)		4.6.29.3	X		
Resistance (hot)		4.6.5			
Insulation resistance (hot)		4.6.30	X		
Efficiency at 4/4, 3/4, 2/4 and 1/4 load		4.6.31	X		
Harmonic analysis		4.6.32	X		
Amplitude modulation		4.6.33	X		
Line unbalance (3 phase generator only)		4.6.34	X		
Voltage balance (3 phase generator only)		4.6.35	X		
Component ratings	4.6.36			X	X
Repair part		4.6.37			
Overvoltage limiting	4.6.38	4.6.38	X		

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4.6.8.1.1, lines 3 and 4: Delete "Specification MIL-E-22843" and substitute "MIL-STD-740".

4.6.8.2.1, lines 4 and 5: Delete "Specification MIL-E-22843" and substitute "MIL-STD-740".

4.6.8.2.2, line 4: Delete "Specification MIL-E-22843" and substitute "MIL-STD-740".

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4.6.13.2.1, lines 4 and 5: Delete "Specification MIL-E-22843" and substitute "MIL-STD-740".

4.6.13.2.2, lines 4 and 5: Delete "Specification MIL-E-22843" and substitute "MIL-STD-740".

PAGE 34

4.6.18, item (b)(4), line 8: Delete "4.6.8" and substitute "4.6.18(b)(10)".

PAGE 35

* 4.6.18, item (d), line 1: Delete "stock" and substitute "shock".

4.6.19, item (a), line 1: Delete "rated frequency" and substitute "nominal voltage".

PAGE 37

4.6.26: Add: "The motor generators shall be paralleled 180 degrees out of phase to demonstrate compliance with 3.2.28(b)".

PAGES 37 and 38

4.6.27: Delete and substitute:

"4.6.27 Short circuit characteristics. The sustained short circuit current shall be measured by means of an ammeter and oscillograph to determine compliance with 3.2.27. When computing the I^2t value the steady state current measured with the ammeter shall be used, the time shall start when the current reaches its steady state value. When the circuit breakers are furnished by the contractor, the test shall be conducted with the circuit breaker locked in to prevent tripping during the test. There shall be no evidence of smoking, burning or burning odor as a result of this test. The voltage shall automatically return to within the regulation band limits when the short circuit is removed. Oscillograms of the fault current and voltage shall be taken."

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PAGE 40

Add new paragraph 4.6.38:

"4.6.38 Overvoltage limitation. Overvoltage limitation features shall be tested as follows: A component failure shall be simulated (by adjusting the rheostat, opening the sensing leads, opening a control or bias lead, or some other means) to show that the voltage will be limited to a value within the overvoltage range. Voltmeter readings and oscillograms shall be taken of the output voltage to determine compliance with 3.5.4.8(f). The motor generator shall be operating at nominal input voltage rated output voltage and frequency and 50 percent rated load before the test. There shall be no damage to the equipment as a result of these tests."

5.1, line 5: Delete "MIL-E-22843" and substitute "MIL-E-16298".

5.5, line 2: Delete "MIL-D-963" and substitute "DOD-D-1000".

PAGE 41

6.1, item (i): Delete.

NOTE: The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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